

Lab 2.3: Implementation of NFA to identify strings having 0011 as Substring

Background and Requirements

In Formal Language Theory and Automata Theory, a **Non-Deterministic Finite Automaton (NFA)** is a finite-state machine that can transition to multiple possible states for a given input symbol. Unlike a Deterministic Finite Automaton (DFA), an NFA may have more than one valid transition from a state, allowing it to explore several computation paths simultaneously. A string is accepted if at least one of these paths ends in an accepting state after processing the entire input.

NFAs are widely used in pattern matching, regular expression processing, lexical analysis, and compiler design because they provide a flexible and intuitive way to recognize complex patterns in input strings.

The task is to accept a string consisting of the symbols '0' and '1' and determine whether it contains the substring "0011".

Source Code

```
alphabet = {"0", "1"}

# Transition table
transition = {
    "q0": {"0": {"q0", "q1"}, "1": {"q0"}},
    "q1": {"0": {"q2"}},
    "q2": {"1": {"q3"}},
    "q3": {"1": {"qf"}},
    "qf": {"0": {"q0", "q1", "qf"}, "1": {"q0", "qf"}},
}

s = input("Enter a string sequence: ")
states = {"q0"}

for ch in s:
    if ch not in alphabet:
        print("Invalid input")
        break

    next_states = set()

    for state in states:
        next_states.update(transition[state].get(ch, set()))

    states = next_states
    print(f"Input: {ch}, Current States: {states}")
else:
    if "qf" in states:
        print("Accepted")
    else:
        print("Rejected")
```

Output

```
Enter a string sequence: 001001101
Input: 0, Current States: {'q0', 'q1'}
Input: 0, Current States: {'q2', 'q0', 'q1'}
Input: 1, Current States: {'q3', 'q0'}
Input: 0, Current States: {'q0', 'q1'}
Input: 0, Current States: {'q2', 'q0', 'q1'}
Input: 1, Current States: {'q3', 'q0'}
Input: 1, Current States: {'qf', 'q0'}
Input: 0, Current States: {'q0', 'q1', 'qf'}
Input: 1, Current States: {'qf', 'q0'}
Accepted
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

```
Enter a string sequence: 10101011
Input: 1, Current States: {'q0'}
Input: 0, Current States: {'q0', 'q1'}
Input: 1, Current States: {'q0'}
Input: 0, Current States: {'q0', 'q1'}
Input: 1, Current States: {'q0'}
Input: 0, Current States: {'q0', 'q1'}
Input: 1, Current States: {'q0'}
Input: 1, Current States: {'q0'}
Rejected
○ (base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works>
```

Conclusion

This lab demonstrated the implementation of a Non-Deterministic Finite Automaton (NFA) to recognize strings containing the substring "0011". The experiment reinforced the concepts of non-deterministic state transitions, multiple active states, and acceptance based on reaching a final state, highlighting the effectiveness of NFAs in pattern recognition.